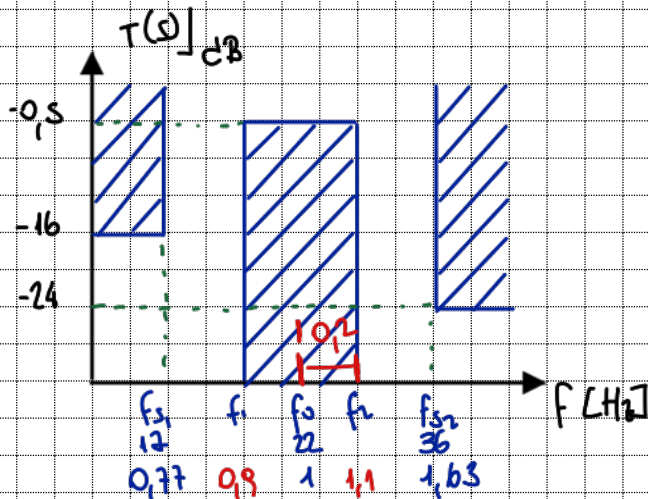


Tarea 4 - Montenegro Maximiliano



$$f_0 = 24 \text{ kHz}$$

$$Q = 5$$

$$f_{s1} = 17 \text{ kHz}$$

$$\omega_1 \omega_2 = \omega_0^2$$

$$f_{s2} = 36 \text{ kHz}$$

$$\omega_2 - \omega_1 = BW = \frac{\omega_0}{Q}$$

Normalizo dividiendo por f_0

$$\Rightarrow BW = \frac{f_0}{Q} = \frac{1}{5} = 0.2$$

$$\omega_2 - \omega_1 = BW$$

$$2\pi f_2 - 2\pi f_1 = 2\pi \frac{f_0}{Q}$$

$$f_2 - f_1 = f_0/Q$$

$$\omega_2 \omega_1 = \omega_0^2$$

$$2\pi f_2 2\pi f_1 = (2\pi)^2 f_0^2$$

$$f_2 f_1 = f_0^2$$

$$\left. \begin{array}{l} f_2 - f_1 = f_0/Q \\ f_2 f_1 = f_0^2 \end{array} \right\} \Rightarrow \left. \begin{array}{l} f_2 - f_1 = 0.2 \\ f_2 f_1 = 1 \end{array} \right\} \Rightarrow \left. \begin{array}{l} f_2 = 0.2 + f_1 \\ (0.2 + f_1) f_1 = 1 \end{array} \right\}$$

$$f_2 f_1 = f_0^2$$

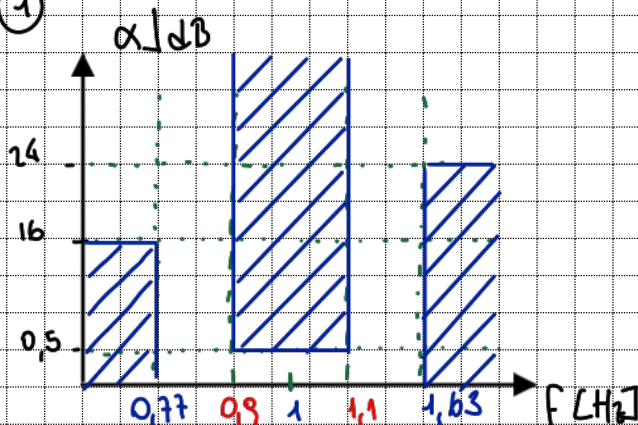
$$f_1 + f_1^2 = 1 \Rightarrow f_1^2 + 0.2f_1 - 1 = 0$$

$$f_1 = 0.9$$

$$\Rightarrow f_2 = 0.2 + 0.9 = 1.1$$

$$f_1 = -1.6183$$

①



$$\alpha_{\max} = 0.5$$

$$\alpha_{\min 1} = 16 \text{ dB}$$

$$\alpha_{\min 2} = 24 \text{ dB}$$

$$10^{\alpha_{\max}/10} - 1 = \epsilon^2 = 0.122$$

②

$$\Omega = \frac{1}{2W} \frac{(\omega^2 - \omega_0^2)}{\omega}$$

$$\Omega \Big|_{\omega=1,63} = \frac{1}{0,2} \frac{(1,63^2 - 1^2)}{1,63} = 5,13$$

$$\Omega \Big|_{\omega=0,77} = \frac{1}{0,2} \frac{(0,77^2 - 1^2)}{0,77} = -2,61$$

$$\alpha_{\min} = 10 \log(1 + \epsilon^2 \cosh^2(n \cosh^{-1}(\Omega)))$$

$$n \quad \alpha_{\min} = 24 \quad |\Omega| = 5,13$$

$$1 \quad 6,24$$

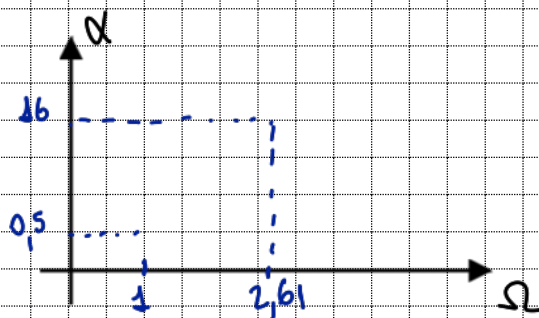
$$2 \quad 25,14$$

$$n \quad \alpha_{\min} = 16 \quad |\Omega| = 2,61$$

$$1 \quad 2,62$$

$$2 \quad 13,11$$

$$3 \quad 26,89$$



se elige la de orden 3, debido a que es la más exigida.

$\therefore$ se realiza un filtro chebyshev de orden 3.

$$C_0 = 1; \quad C_1 = W \quad C_n = 2WC_{n-1} - C_{n-2}$$

$$C_2 = 2W^2 - 1 = 2W^2 - 1$$

$$C_3 = 2W(2W^2 - 1) - W = 4W^3 - 2W - W = 4W^3 - 3W$$

$$T(s)T(-s) = \frac{1}{1 + \epsilon^2 (C_3(W))^2}$$

$$= \frac{1}{1 + \epsilon^2 (4W^3 - 3W)^2}$$

$$T(s)(-s) = \frac{1}{1 + \xi^2 16 \omega^6 - 24 \xi^2 \omega^4 + 9 \xi^2 \omega^2} \quad s = j\omega \Rightarrow \omega = \frac{1}{j}s$$

$$= \frac{1}{1 + \xi^2 16 (-s^6) - 24 \xi^2 (s^4) + 9 \xi^2 (-s^2)} \quad \therefore \omega = -js \Rightarrow \omega^2 = -s^2$$

$$= \frac{1}{1 - \xi^2 16 s^6 - 24 \xi^2 s^4 - 9 \xi^2 s^2} \quad \omega^4 = s^4$$

$$\omega^6 = -s^6$$

$$T(s)T(-s) = \frac{1}{\xi^2 16} \cdot \frac{1}{s^6 + \frac{24}{16} s^4 + \frac{9}{16} s^2} = \frac{1}{\xi^2 16}$$

Mediante el scrip de python se obtiene:

$$T(s) = \frac{0,7157}{s^3 + 1,253 s^2 + 1,535 s + 0,7157}$$

$$T(s) = \frac{0,6264}{s + 0,6264} \cdot \frac{1,069^2}{s^2 + \frac{1,069}{1,706} s + 1,069^2}$$

$$\omega_0 = 1,069$$

$$Q = 1,706$$

③

Pasa bajo

Pasa banda

$$s = Q \frac{s^2 + 1}{s}$$

$$\frac{\theta}{s + \theta}$$

$$\frac{\theta}{Q \frac{s^2 + 1}{s} + \theta}$$

$$\frac{s}{Q s^2 + Q + \theta s} = \frac{1/Q s}{s^2 + s \theta/Q + 1}$$

$$\frac{b}{s^2 + sa + b}$$

$$\frac{b}{\left(Q \frac{s^2 + 1}{s}\right)^2 + \left(Q \frac{s^2 + 1}{s}\right)a + b}$$

$$\frac{b s^2}{(s^4 + 2s^2 + 1)Q^2 + aQ(s^2 + 1)s + b s^2} = \frac{b s^2}{s^4 Q^2 + s^3 aQ + s^2(2Q^2 + b^2) + s aQ + Q^2}$$

$$\frac{b/Q^2 s}{s^4 + s^3 a/Q + s^2(2 + b^2/Q^2) + s a/Q + 1} = \frac{s \omega_0/Q_1 k_1}{s^2 + s \frac{\omega_0}{Q_1} + \omega_0^2} + \frac{s \omega_0/Q_2 k_2}{s^2 + s \frac{\omega_0}{Q_2} + \omega_0^2}$$

$$\Rightarrow P_b \ n=3 \rightarrow P_B \ n=5$$

$$Pasa \ bajo \quad n=3 \quad Q=5$$

$$T(s) = \frac{0,6264}{s+0,6264} \cdot \frac{1,069^2}{s^2 + \frac{1,069}{1,706} s + 1,069^2} \quad \begin{matrix} \omega_0 = 1,069 \\ Q = 1,706 \end{matrix}$$

$$\textcircled{1} \quad \frac{0,6264}{s+0,6264} \rightarrow \frac{0,6264}{Q(s^2+1)+0,6264} = \frac{0,6264 \cdot s}{5s^2+5+s \cdot 0,6264}$$

$$\frac{0,6264/5 \cdot s}{s^2 + s \frac{0,6264}{5} + 1} = 0,125 \frac{s}{s^2 + s \cdot 0,125 + 1}$$

$$\textcircled{2} \quad \frac{1,069^2}{s^2 + 0,6266 s + 1,069^2} \rightarrow \frac{s^2 \cdot 0,1069^2/5^2}{s^4 + s^3 \frac{0,6266}{5} + s^2(2 + \frac{1,069^2}{5^2}) + s \frac{0,6266}{5} + 1}$$

$$\frac{s^2 \cdot 0,046}{s^4 + s^3 \cdot 0,125 + s^2 \cdot 2,046 + s \cdot 0,125 + 1}$$

$$\begin{cases} Q_1 = Q_2 = Q & Q = 2Q \cdot Q_{PB} \\ \omega_{01} = \omega_{02} = 1^2 & \omega_2 = 1 + \frac{1}{2Q} \Rightarrow \omega_1 = \frac{1}{\omega_2} \end{cases}$$

$$q = 2 \cdot 5 \cdot 1,706 = 17,06$$

$$\omega_2 = 1 + \frac{1}{2(17,06)} \approx 1,029 \Rightarrow \frac{s \cdot 1,029/17,06}{s^2 + s \frac{1,029}{17,06} + 1,029^2}$$

$$\omega_1 = \frac{1}{1,029} \approx 0,972 \Rightarrow \frac{s \cdot 0,972/17,06}{s^2 + s \frac{0,972}{17,06} + 0,972^2}$$

$\therefore$ la transferencia obtenida es la siguiente

$$T(s) = \frac{0,125 s}{s^2 + s \cdot 0,125 + 1} \cdot \frac{s \cdot 0,057}{s^2 + s \cdot 0,057 + 0,944} \cdot \frac{s \cdot 0,060}{s^2 + s \cdot 0,060 + 1,058}$$

Pasabanda total

$$T(s) = \frac{0,7157}{s^3 + 1,253 s^2 + 1,535 s + 0,7157}$$

$f = Q \left(\frac{s^2+1}{s} \right)$
 $f = 5 \left(\frac{s^2+1}{s} \right)$

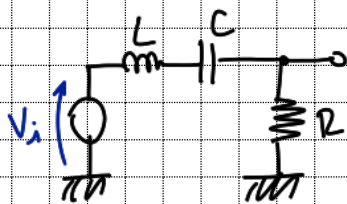
$$\frac{0,7157}{\left(5 \left(\frac{s^2+1}{s} \right) + 1,253 \left(5 \left(\frac{s^2+1}{s} \right) \right)^2 + 1,535 s \left(\frac{s^2+1}{s} \right) + 0,7157 \right)}$$

$$\frac{0,7157}{125 \left(\frac{s^6+3s^4+3s^2+1}{s^3} \right) + 1,253 \left(\frac{25(s^4+2s^2+1)}{s^2} \right) + 1,535 \left(\frac{s^3+s}{s} \right) + 0,7157}$$

$$\frac{0,7157 s^3}{125 (s^6 + 3s^4 + 3s^2 + 1) + 1,253 \cdot 25 (s^5 + 2s^3 + s) + 1,535 \cdot 5 (s^4 + s^2) + 0,7157 s^3}$$

$$\frac{0,7157/125 s^3}{s^6 + 3s^4 + 3s^2 + 1 + \frac{1,253}{5} (s^5 + 2s^3 + s) + \frac{1,535}{25} (s^4 + s^2) + \frac{0,7157}{125} s^3}$$

$$T(s) = \frac{s^3 \cdot 0,005726}{s^6 + s^5 0,251 + s^4 3,061 + s^3 0,507 + s^2 3,061 + s 0,251 + 1}$$



$$I(s) = \frac{V_i(s)}{Z(s)}$$

$$Z(s) = R + sL + \frac{1}{sC}$$

$$Z(s) = \frac{s^2 RC + s^2 LC + 1}{sC}$$

$$V_L(s) = I(s) \cdot R = \frac{V_i(s)}{\frac{s^2 LC + s^2 RC + 1}{sC}} R$$

$$\frac{V_L(s)}{V_i(s)} \cdot T(s) = \frac{sRC}{s^2 LC + s^2 RC + 1} = \frac{RL}{LC} \frac{s}{s^2 + \frac{sR}{L} + \frac{1}{LC}}$$

$$T(s) = \frac{s R/L}{s^2 + \frac{sR}{L} + \frac{1}{LC}}$$

$$\frac{1}{LC} = \omega_0^2 \quad \eta_L = \frac{\omega_0}{Q}$$

$$T(s) = \frac{s \omega_0 / Q}{s^2 + s \omega_0 / Q + \omega_0^2}$$

$$s = s \omega_0$$

$$R=1$$

$$\frac{R}{L} = \frac{\omega_0}{Q}$$

$$L = \frac{Q}{\omega_0}$$

$$T(s) = \frac{s \omega_0 \cdot \omega_0 / Q}{(s \omega_0)^2 + s \omega_0 \omega_0 / Q + \omega_0^2}$$

$$\frac{s \cdot 1/Q}{s^2 + s \cdot \frac{1}{Q} + 1}$$

$$\frac{1}{LC} = \omega_0^2$$

$$C = \frac{1}{\omega_0 Q}$$

Parámetros para el Spice

$$T(s) = \frac{0,125s}{s^2 + s0,125 + 1} \cdot \frac{s}{s^2 + s0,057 + 0,944} \cdot \frac{s}{s^2 + s0,060 + 1,058}$$

$$\bullet \quad \omega_{01}^2 = 1 \Rightarrow \omega_{01} = 1$$

$$\frac{\omega_{01}}{Q_1} = 0,125 \Rightarrow Q_1 = \frac{\omega_{01}}{0,125}$$

$$\bullet \quad \omega_{02}^2 = 0,944 \Rightarrow \omega_{02} = \sqrt{0,944} \quad Q_2 = \frac{\omega_{02}}{0,057}$$

$$\bullet \quad \omega_{03}^2 = 1,058 \Rightarrow \omega_{03} = \sqrt{1,058} \quad Q_3 = \frac{\omega_{03}}{0,06}$$

f pasa bajos $\rightarrow$ f pasa alto

$$s = \frac{1}{S}$$

$$\Rightarrow T(s) = \frac{0,7157}{s^3 + 1,253 s^2 + 1,535 s + 0,7157}$$

$P_b \rightarrow P_A$ ↓

$$T(s) = \frac{0,7157}{\left(\frac{1}{S}\right)^3 + 1,253 \left(\frac{1}{S}\right)^2 + 1,535 \frac{1}{S} + 0,7157}$$

$$T(s) = \frac{0,7157 s^3}{0,7157 s^3 + 1,535 s^2 + 1,253 s + 1}$$

$$T(s) = \frac{s^3}{s^3 + \frac{1,535}{0,7157} s^2 + \frac{1,253}{0,7157} s + \frac{1}{0,7157}}$$

$$T(s) = \frac{s^3}{s^3 + 2,145 s^2 + 1,750 s + 1,397}$$